EE – 1118 LAB - 4

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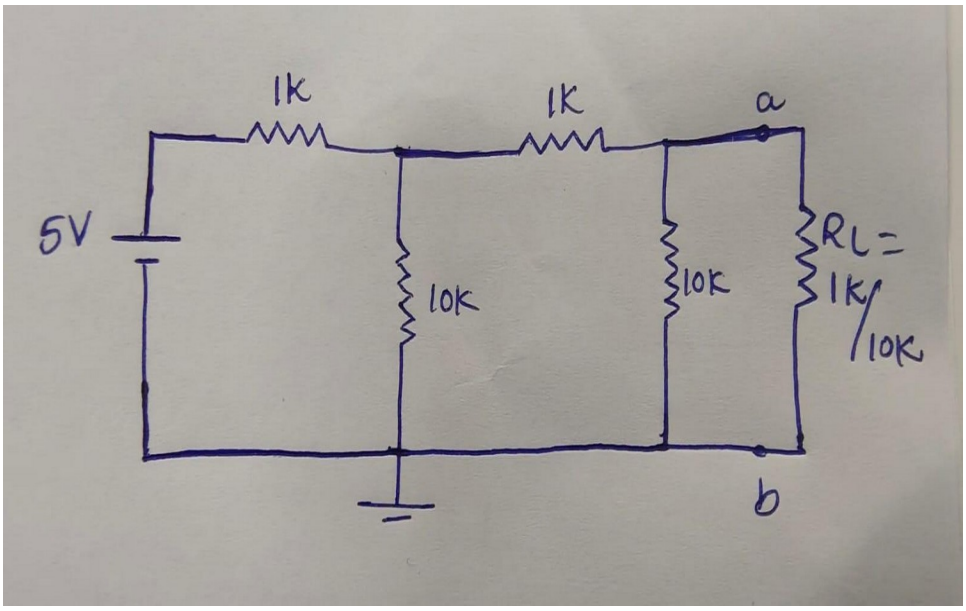
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Section: A-1

Practical demonstration of Thevenin's Theorem applied to a resistive network.

1. Objective:

To confirm Thevenin's Theorem by analyzing the given resistive network and comparing calculated results with measured ones.



2. Apparatus Required:

- DC power source (5 V)
- Resistors — 1 kΩ (×2), 10 kΩ
- Load resistors — 1 kΩ and 10 kΩ
- Digital multimeter / voltmeter / ammeter
- Breadboard and jumper wires

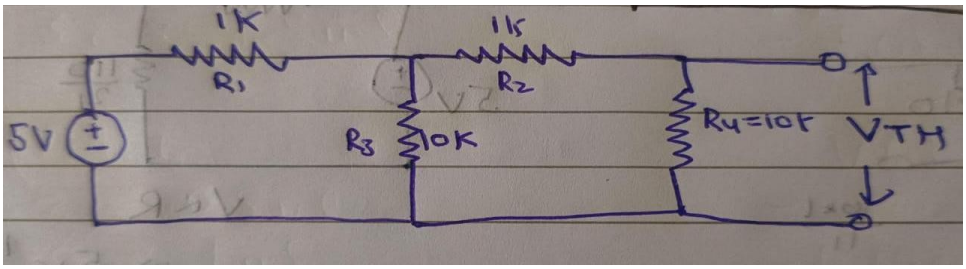
3. Theory:

The current through the load can be found using:

IL = VTH / (RTH + RL)

4. Calculations:

Step 1: Determining VTH



Applying KVL to Loop 1:

$-5 + 1 \cdot I1 + 10 \cdot I1 - 10 \cdot I2 = 0$

$11 \cdot I1 - 10 \cdot I2 = 5$ ...(1)

Applying KVL to Loop 2:

$10 \cdot I2 - 10 \cdot I1 + 1 \cdot I2 + 10 \cdot I2 = 0$

$-10 \cdot I1 + 231 \cdot I2 = 0$ ...(2)

Scaling equation (1) by 10 and equation (2) by 11, then adding the results:

$110 \cdot I1 - 100 \cdot I2 = 50$

$-110 \cdot I1 + 231 \cdot I2 = 0$

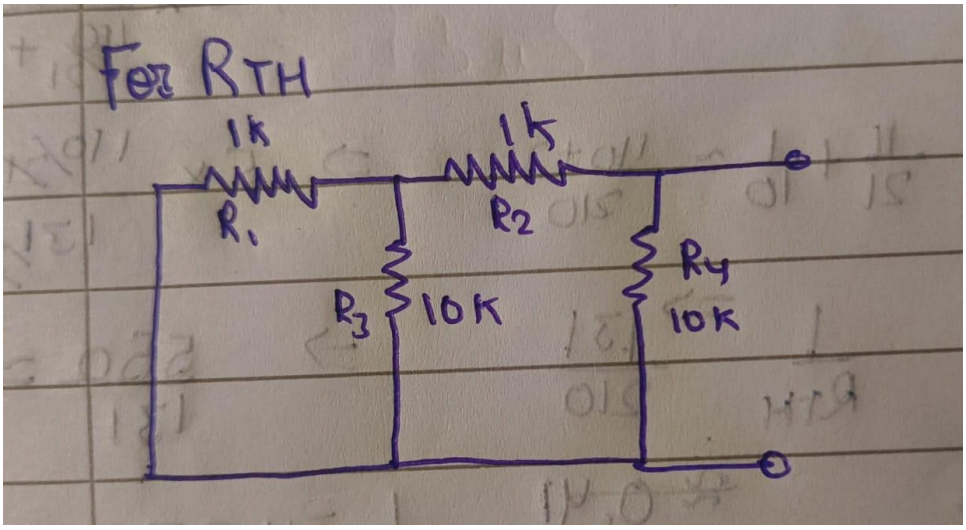
Adding: $131 \cdot I2 = 50$

$I2 = 50/131 = 0.38168$ mA

$VTH = I2 \times 10\ k\Omega = 0.38168\ mA \times 10\ k\Omega = 3.8168\ V \approx 3.82\ V$

Step 2: Determining RTH

To obtain RTH, deactivate the source (short the 5 V supply) and disconnect the load, then compute the resistance seen between terminals a and b.



$RTH = (21/11 \times 10) / (21/11 + 10)$

$RTH = 210/131\ k\Omega = 1.6031\ k\Omega \approx 1.60\ k\Omega$

Case 1: RL = 1 kΩ

$IL = VTH / (RTH + RL) = 3.8168 / (1.60 + 1) = 3.8168 / 2.60$

IL = 1.468 mA ≈ 1.46 mA

Case 2: RL = 10 kΩ

IL = VTH / (RTH + RL) = 3.8168 / (1.60 + 10) = 3.8168 / 11.60

IL = 0.3290 mA ≈ 0.329 mA

5. Observation Table

Percentage Error = [(Theoretical − Practical) / Theoretical] × 100

	Theoretical Values			Practical Values			% Error		
RL (kΩ)	**VTH**	**RTH**	**IL(mA)**	**VTH**	**RTH**	**IL(mA)**	**VTH**	**RTH**	**IL**
1	3.82	1.60	1.46	3.82	1.58	1.464	0.08%	1.44%	0.16%
10	3.82	1.60	0.329	3.82	1.58	0.342	0.08%	1.44%	3.97%

6. Conclusion:

The measured values closely match the calculated results, confirming that Thevenin's Theorem holds true for the tested circuit.